

is solved first using a second-order explicit Taylor–Galerkin method. This is followed by a viscous diffusion phase solved using the forward Euler method. Finally, the pressure/incompressibility phase is solved using the backward Euler scheme. The present results can also be compared with those obtained by Blasco, Codina and Huerta (1997) using a predictor-multicorrector algorithm.

6.8.4 Natural convection in a square cavity

To illustrate the use of the Chorin–Temam fractional-step method in another incompressible flow problem, we shall describe the splitting-up approximate solution of unsteady natural convection problems. Such problems involve a coupling between the Navier–Stokes equations describing the fluid motion and the thermal energy equation governing the space–time evolution of the temperature. The forces which induce natural convection are in fact spatially variable gravity forces generated by (buoyancy) density variations in the fluid due to the non-uniformity of the temperature.

6.8.4.1 The Navier–Stokes and temperature equations In natural convection problems the mathematical description of fluid motion is given by the following form of the Navier–Stokes equations:

$$\varrho(\mathbf{v}_t + (\mathbf{v} \cdot \nabla)\mathbf{v}) = \nabla \cdot \boldsymbol{\sigma} + \varrho \mathbf{b} \quad \text{in } \Omega \times]0, T[, \quad (6.46)$$

where

$$\mathbf{b} = (1 - \beta(T - T_0)) \mathbf{g} \quad (6.47)$$

is the gravity force per unit mass derived on the basis of Boussinesq approximation, vector $\mathbf{g}$ denotes the gravity field, β is the coefficient of thermal expansion of the fluid, T is the temperature field, T_0 is the reference temperature, and ϱ is the reference density of the fluid (i.e., the fluid density corresponding to the reference temperature, when $T = T_0$).

The fluid is assumed incompressible according to Boussinesq approximation. That is, its density is assumed constant, except in the gravity force term where it depends on temperature according to the indicated linear law, see (6.47).

Under the usual assumption of a Newtonian fluid, the components σ_{ij} of Cauchy stress are linearly related to the strain rate by Stokes' law. The incompressibility condition is expressed in the standard form $\nabla \cdot \mathbf{v} = 0$, and the Navier–Stokes equations (6.46) are completed with the usual initial and boundary conditions.

The Boussinesq body forces, see (6.47), introduce a coupling between the Navier–Stokes equations and the thermal energy equation

$$\varrho c(T_t + \mathbf{v} \cdot \nabla T) = \nabla \cdot (k \nabla T) + Q \quad \text{in } \Omega \times]0, T[. \quad (6.48)$$

Here, c denotes the specific heat of the fluid at constant volume, k is the thermal conductivity (assumed isotropic in this case) and Q represents volumetric heat generation.

Equation (6.48) combined with boundary conditions such as

$$\begin{aligned} T(\mathbf{x}, t) &= T_D(\mathbf{x}, t) && \text{on } \Gamma_D^T \times]0, T[, \\ k \frac{\partial T(\mathbf{x}, t)}{\partial n} &= q_N(\mathbf{x}, t) && \text{on } \Gamma_N^T \times]0, T[, \end{aligned}$$

where T_D represents a prescribed temperature, while q_N denotes a prescribed normal heat flux. The initial condition $T(\mathbf{x}, 0) = T_0(\mathbf{x})$ for all $\mathbf{x} \in \Omega$, completes the initial boundary value problem for the temperature.

6.8.4.2 A fractional-step method for Navier–Stokes and temperature

The implicit character of pressure in an incompressible fluid precludes the use of purely explicit time integration algorithms in the Navier–Stokes equations. As seen in Section 6.7.3, it is nevertheless possible to take advantage of the algorithmic simplicity of explicit time-stepping schemes by using a fractional-step approach. That is, the convective and Boussinesq terms could be evaluated explicitly. Here a simple example is shown and to simplify the exposition, we assume that Dirichlet conditions for the velocity are prescribed on the whole boundary of the fluid domain.

The first step considers the convective, viscous and Boussinesq terms of equation (6.46), see also (6.37). Thus, the first-step equation reads

$$\mathbf{v}_t + (\mathbf{v} \cdot \nabla) \mathbf{v} - \nabla \cdot (2\nu \nabla^S \mathbf{v}) = \mathbf{b}.$$

Note that this equation is normalized by the reference density, ρ , of the fluid. The simplest option for this first step is an explicit time integration scheme. For instance, we can use the explicit Euler method or the second-order Adams–Bashforth method described in Section 5.3.1. The result is an intermediate velocity field $\mathbf{v}_{\text{int}}^{n+1}$, satisfying the Dirichlet boundary conditions, that is $\mathbf{v}_{\text{int}}^{n+1} = \mathbf{v}_D^{n+1}$.

The second step of the time integration of the Navier–Stokes equations is the pressure/incompressibility step. The end-of-step velocity field $\mathbf{v}^{n+1}$ is determined adding the intermediate velocity $\mathbf{v}_{\text{int}}^{n+1}$ and the effect of the pressure gradient. Recall that pressure is such that the end-of-step velocity is solenoidal. Time integration is necessarily implicit and the simplest algorithm is given by the backward Euler method, see equations (6.40).

A two-step procedure can also be used to update temperature from T^n to T^{n+1} . The convective term is integrated in the first step and the diffusion and source terms in the second one, see equation (6.48). Thus, the first step reduces to the hyperbolic equation

$$T_t + \mathbf{v} \cdot \nabla T = 0.$$

The third-order explicit Taylor–Galerkin method, see Section 3.6.2, can be used for the time integration of this equation. An intermediate temperature field, T_{int} , is then obtained. Since this is a pure convection equation Dirichlet boundary condition $T_{\text{int}} = T_D^{n+1}$ are only prescribed on the inflow portion of Γ_D^T (i.e., where $\mathbf{n} \cdot \mathbf{v} < 0$).

The second step, which accounts for the diffusion and source terms, reads

$$\rho c T_t = \nabla \cdot (k \nabla T) + Q.$$

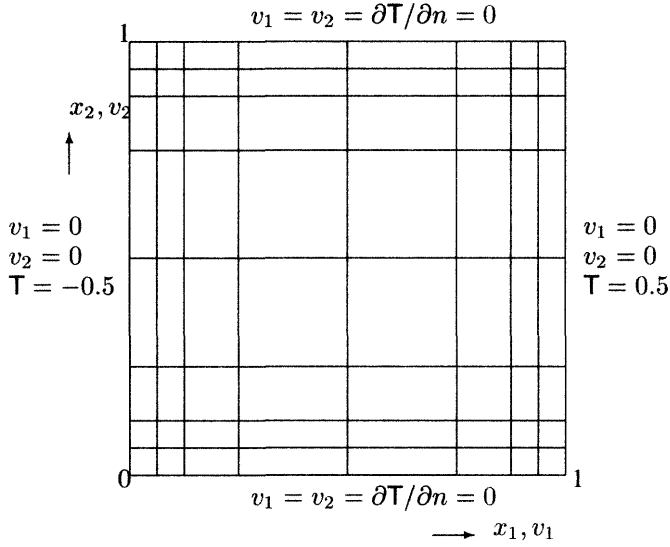


Fig. 6.16 Natural convection: problem statement and discretization.

The complete (Dirichlet and Neumann) boundary conditions are imposed. In this parabolic phase an implicit time-stepping algorithm is to be preferred since the stable time step for explicit methods decreases with the square of the mesh size. For instance, the second-order Crank–Nicolson method can be a reasonable choice.

6.8.4.3 The numerical example The classical test problem of a differentially heated square cavity is chosen to illustrate the use of the fractional-step method for solving natural convection flows in enclosures.

The problem statement is depicted in Figure 6.16. At the initial time, a fluid at uniform temperature $T_0 = 0$ is at rest in a square cavity with unit sides. At the start of the calculation, the temperature of the left wall of the cavity is suddenly lowered by 0.5, while the temperature of the right wall is increased by the same quantity. The horizontal walls are assumed to be thermally insulated (zero heat flux). No-slip and no penetration velocity boundary conditions are prescribed on all walls.

The motion induced by natural convection is governed by two dimensionless numbers, the Rayleigh and the Prandtl numbers defined by

$$Ra = \frac{g\beta L_{\text{ref}}^3 \Delta T}{\nu \kappa} \quad \text{and} \quad Pr = \frac{\nu}{\kappa},$$

where g is the gravitational acceleration, β the coefficient of thermal expansion, ΔT the temperature difference between the hot and cold walls, ν denotes the kinematic viscosity and κ is the thermal diffusivity of the fluid. The reference length is $L_{\text{ref}} = 1$, the length of the cavity side.

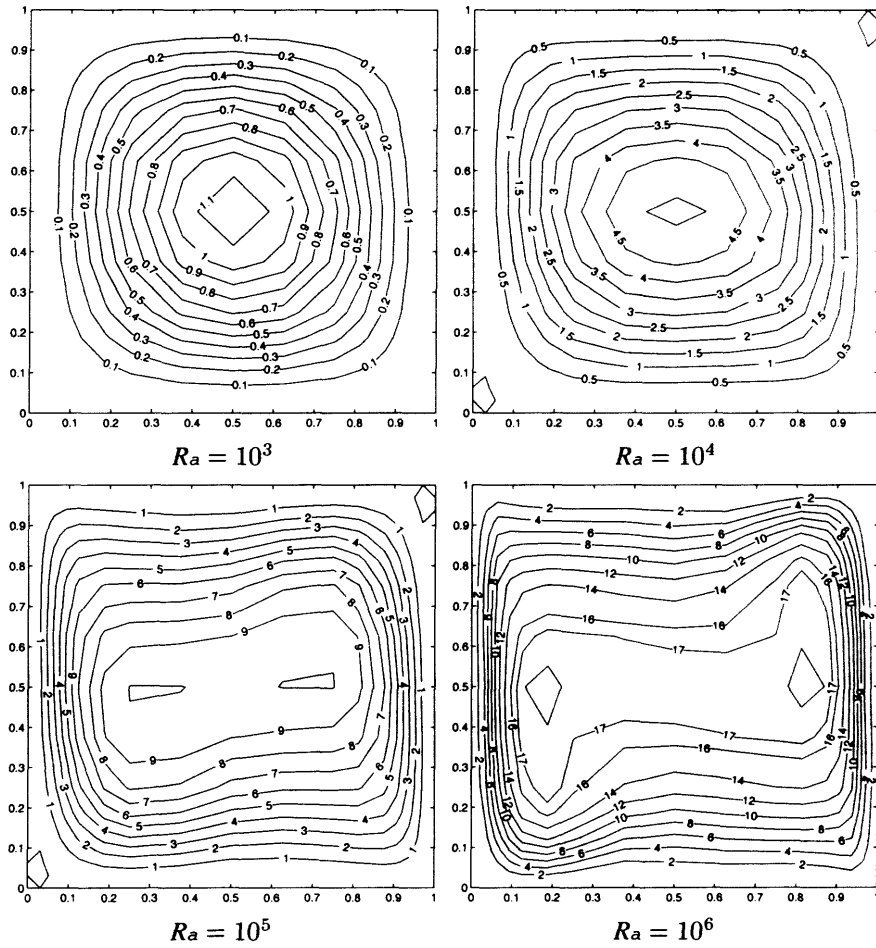


Fig. 6.17 Natural convection: streamlines in the thermal cavity for different Rayleigh numbers and $Pr = 1$.

The problem is symmetric with respect to the center of the cavity. This property can be exploited to reduce the cost of the simulation or to test the quality of the numerical results. We select the second option. The finite element computation is performed using the non-uniform mesh of 8×8 Q2Q1 elements for the mixed interpolation of velocity and pressure, see Figure 6.16. A biquadratic interpolation is used for the temperature. The time integration based upon the fractional-step approach described in Section 6.8.4.2 is used until a steady state is reached.

The results are for a Prandtl number of one and Rayleigh numbers from 10^3 to 10^6 . The development and the structure of the flow vary considerably with the Rayleigh number. The main characteristics of the flow are illustrated in Figures 6.17 and 6.18:

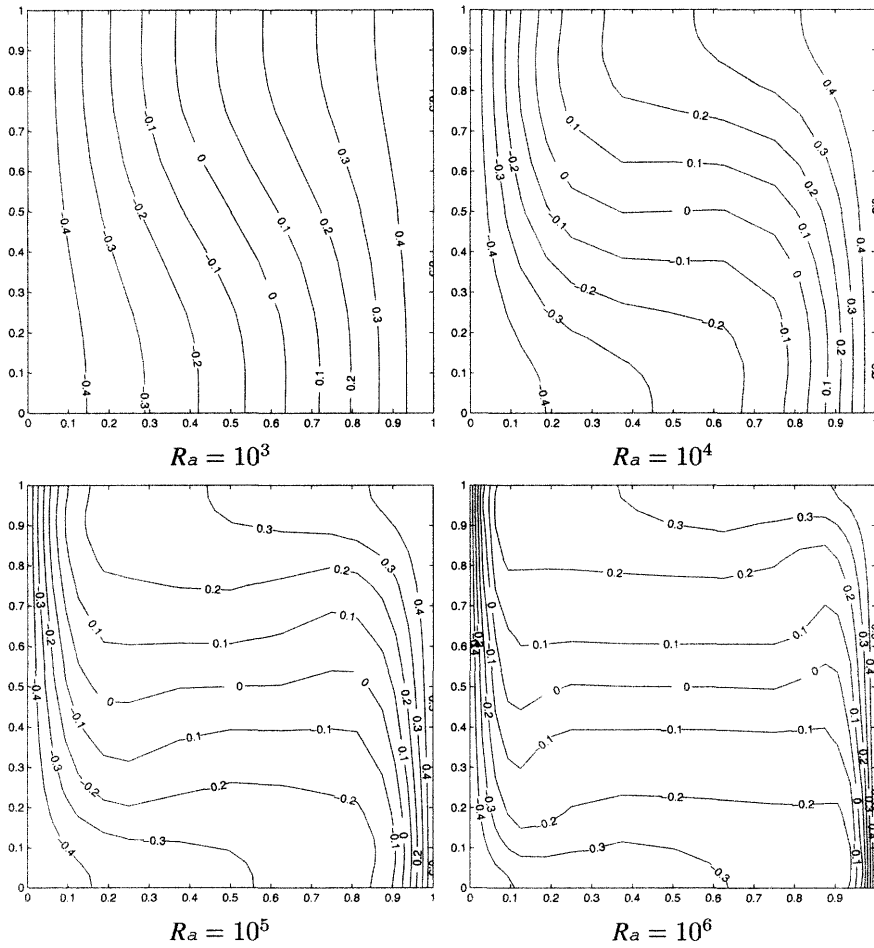


Fig. 6.18 Natural convection: isotherms in the thermal cavity for different Rayleigh numbers and $Pr = 1$.

1. A secondary flow develops at a Rayleigh number situated between 10^4 and 10^5 .
2. A boundary layer appears progressively in the vicinity of the vertical walls of the cavity.
3. The secondary flow evolves at the center of the cavity in the form of “eyes of a cat”.

The computed streamlines are plotted in Figure 6.17. For $Ra < 10^5$, the flow is mono-cellular. The fluid rises along the hot side of the cavity (right wall) and comes down along the cold wall. For $Ra = 10^5$, the development of a secondary flow starts in the central portion of the cavity. The circulation direction of the secondary flow is the same as that of the base flow. The value of the stream function at the center

Table 6.3 Stream function values at the center of the thermal cavity

Rayleigh number		10^3	10^4	10^5	10^6
Present simulation	(8×8)	1.19	5.15	9.74	17.99
Heinrich et al. (1978)	(4×4)	1.18	5.13	9.44	—
Marshall et al. (1978)	(8×8)	—	5.12	9.54	17.32

of the cavity gives a measure of the importance of the recirculation. The results are displayed in Table 6.3 in comparison with other simulations from the literature.

The isotherms in the cavity are displayed in Figure 6.18. At low values of the Rayleigh number they are practically straight lines, in accordance with a diffusion-dominated heat transfer process within the cavity. When the Rayleigh number increases, the convective effects become more and more important. The isotherms deform progressively and take an S shape. The distortion of the isotherms generates high temperature gradients near the vertical walls of the cavity.